



Fuzzy vs SNN Under Noise

Robustness evaluation under a unified corruption severity scale

Why Compare Model Robustness?

The Challenge

Real-world data arrives corrupted through sensor noise, lossy compression, and transmission errors. Understanding how different architectures handle these corruptions is critical for deployment.

1

Model Degradation

How does performance decay as noise severity increases across different corruption types?

Two Architectures

Fuzzy model: RBF + logistic regression
(interpretable)

Shallow neural network: 1-hidden-layer MLP
(learned features)

2

Preprocessing Value

Does lightweight denoising actually improve robustness, or does it introduce new problems?

Unified Severity Scale

Different noise types are mapped to a single severity parameter $s \in [0,1]$, enabling direct comparison of robustness curves across corruption families.

Gaussian Noise

Standard deviation increases
with s

JPEG Compression

Quality factor decreases with s

Motion Blur

Kernel length grows with s

This unified framework allows apples-to-apples comparison of model resilience patterns across fundamentally different corruption mechanisms.

Model Architectures

Fuzzy Model

Centroid-based RBF rules feed into a logistic readout layer. Highly interpretable through explicit feature space partitioning.

Advantage: Transparent decision boundaries

Limitation: Brittle to structural corruption

The fundamental trade-off: interpretability versus robustness. The fuzzy model's explicit structure becomes a vulnerability when that structure collapses.

Shallow Neural Network

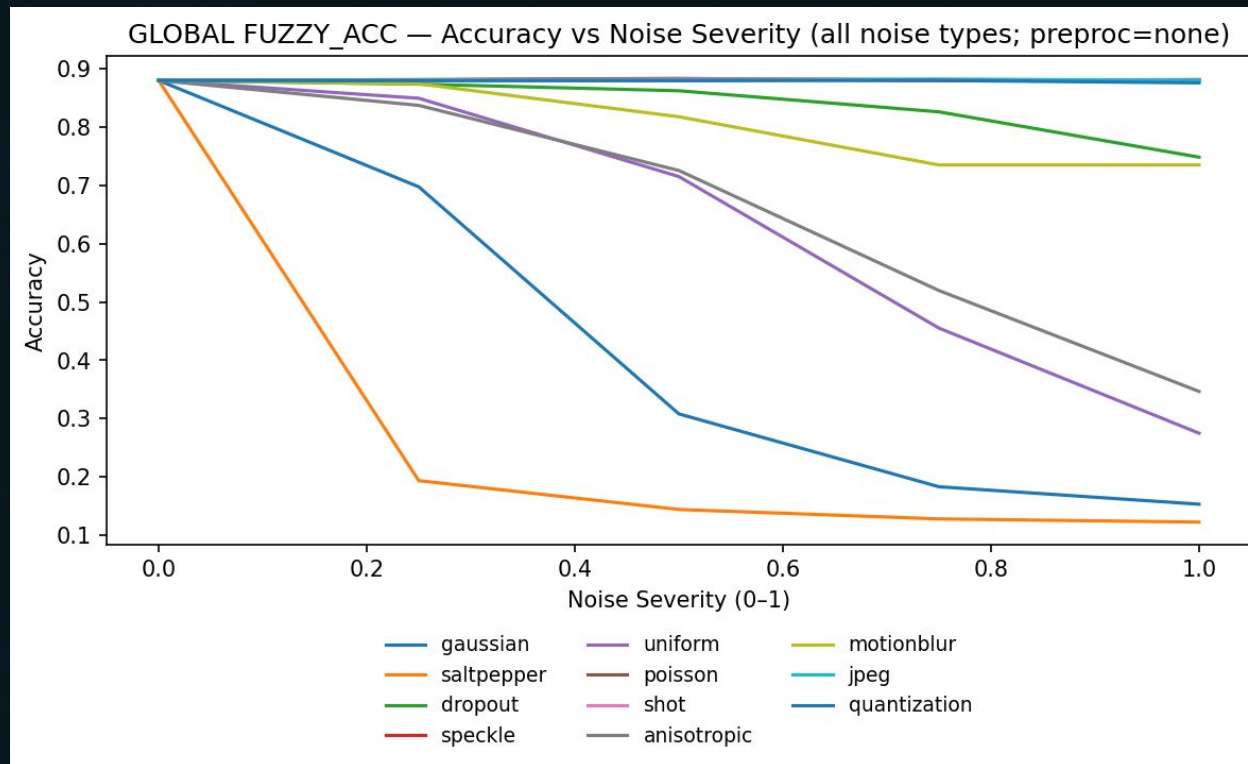
Single hidden layer MLP operates on PCA-reduced features, learning compressed representations during training.

Advantage: Adaptive feature learning

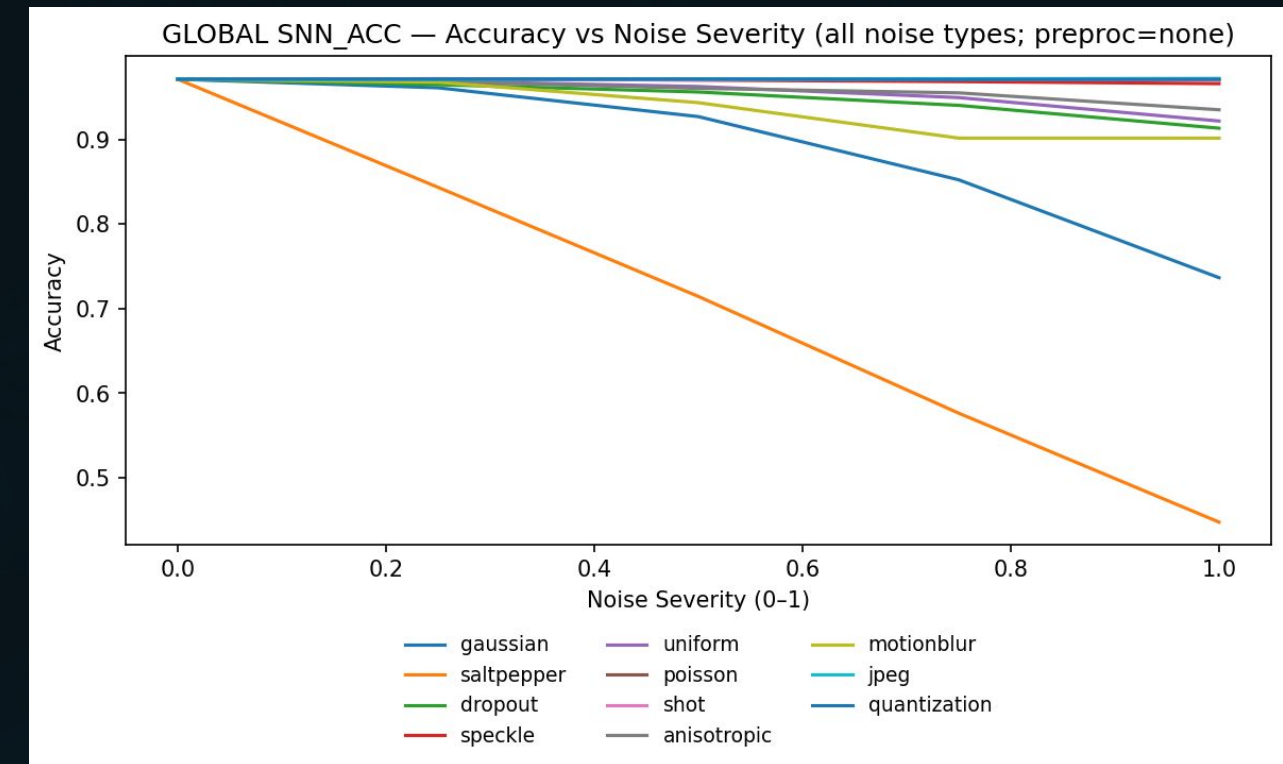
Limitation: Black-box behavior

Baseline performance

Fuzzy model performance



SNN model performance



Baseline: SNN Resilience

The shallow neural network demonstrates substantially greater resilience across all corruption types. Performance degrades gradually rather than collapsing.

Smooth Degradation

Learned representations distribute risk across multiple features, preventing single-point failures

Extreme Noise Limits

Even SNNs fail under severe salt-pepper corruption, but maintain partial functionality longer

Baseline: Fuzzy Model Performance

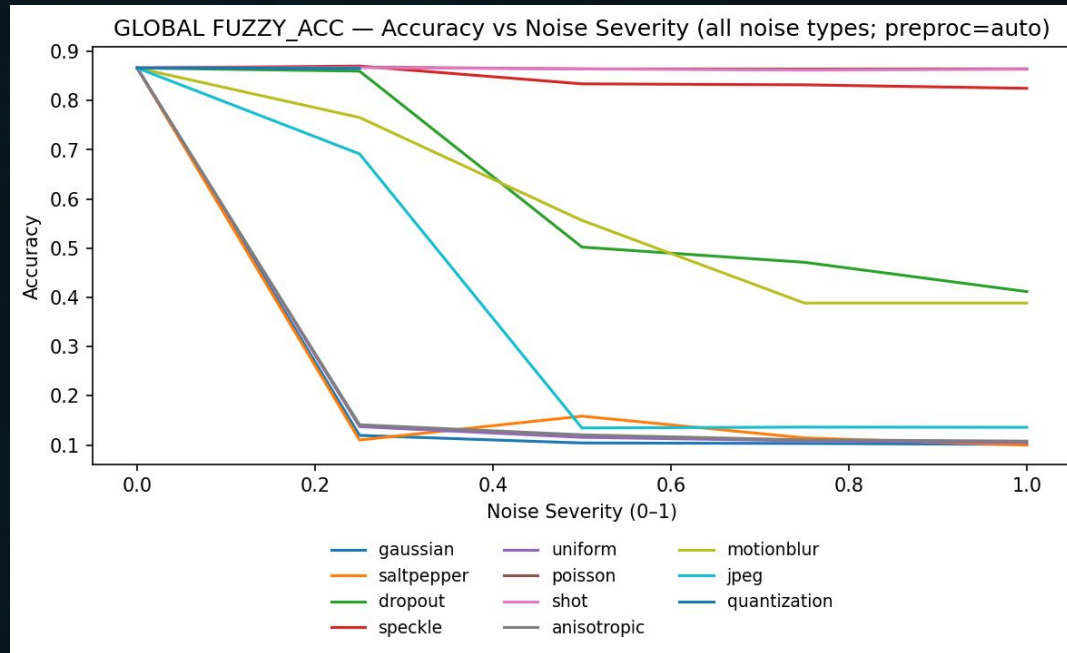
→ **Discontinuous noise (salt-pepper):** Catastrophic failure as centroids lose coherence

→ **Root cause:** When pixel dropout breaks stroke continuity, the centroid-based structure collapses entirely

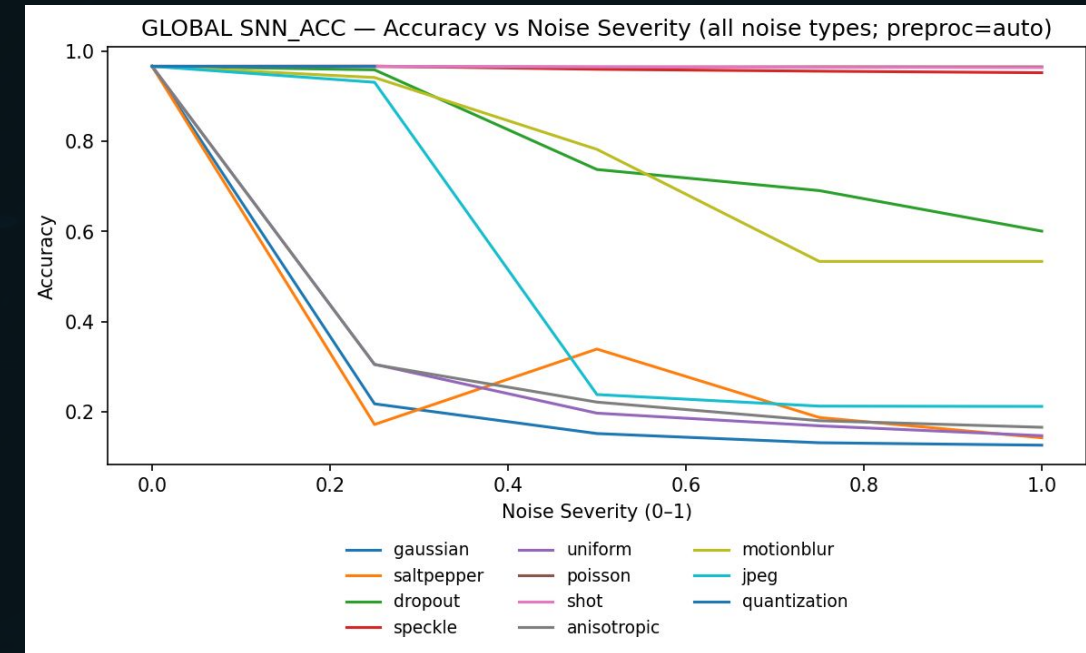
→ **Smooth noise (Gaussian, blur):** Graceful degradation with moderate resilience

First Attempt: Inference-Only Preprocessing

Fuzzy + Denoising



SNN + Denoising



Marginal Gains

Preprocessing provides only slight improvement, primarily at mid-range severities



Train-Test Mismatch

Models trained on clean data struggle with preprocessed test inputs—statistical shift dominates

The Train-Test Mismatch

Problem

The denoiser changes data statistics more than it restores underlying structure. Models trained on pristine strokes now face filtered strokes with altered feature distributions.

1

Clean Training

Model learns decision boundaries on original data manifold

2

Filtered Testing

Preprocessing shifts data into unfamiliar regions of feature space

3

Net Effect

Distribution shift harm exceeds noise reduction benefit

Solution: Train-Time Alignment

Fuzzy with alignment

SNN with alignment

Apply identical preprocessing during both training and inference, with validation-based gating to reject unhelpful filtering.

0

1

Consistency

Same denoising pipeline in training and testing eliminates distribution mismatch

0

2

Validation Gate

Only apply preprocessing when validation performance improves—automatic rejection of harmful filtering

0

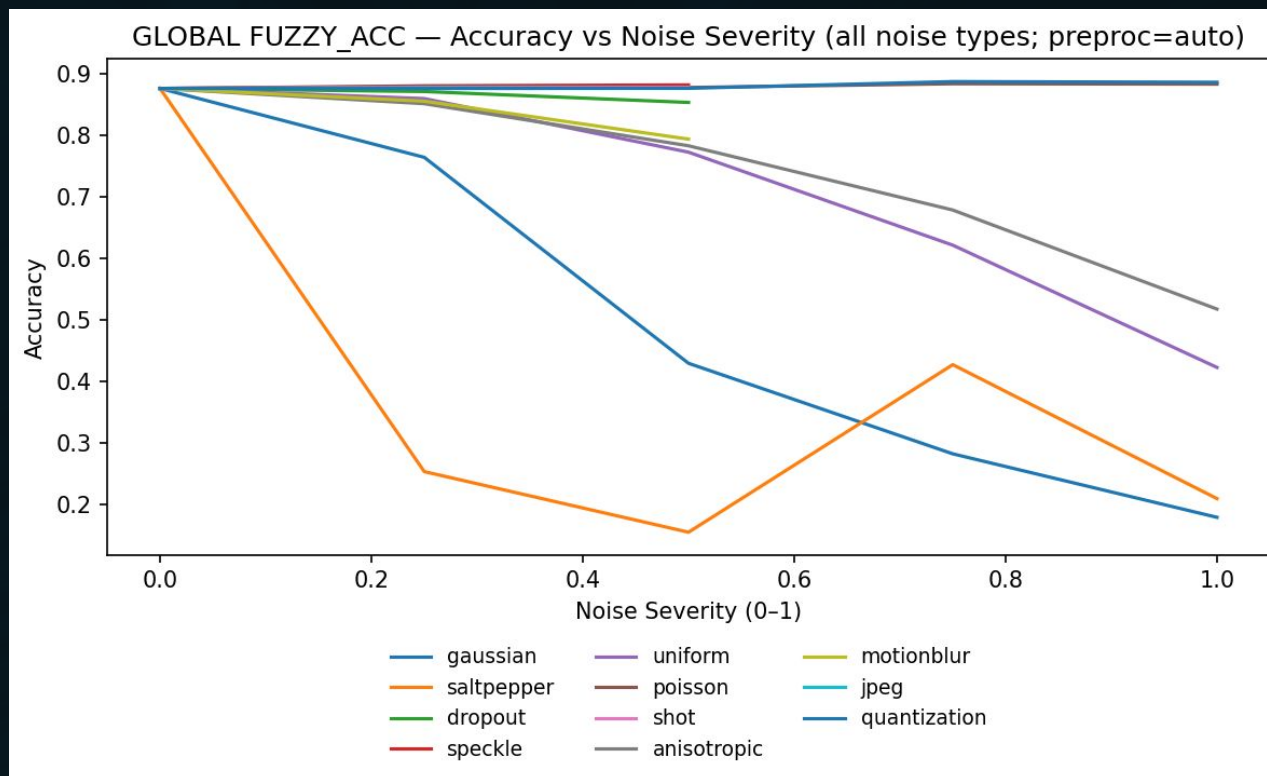
3

Adaptive Curves

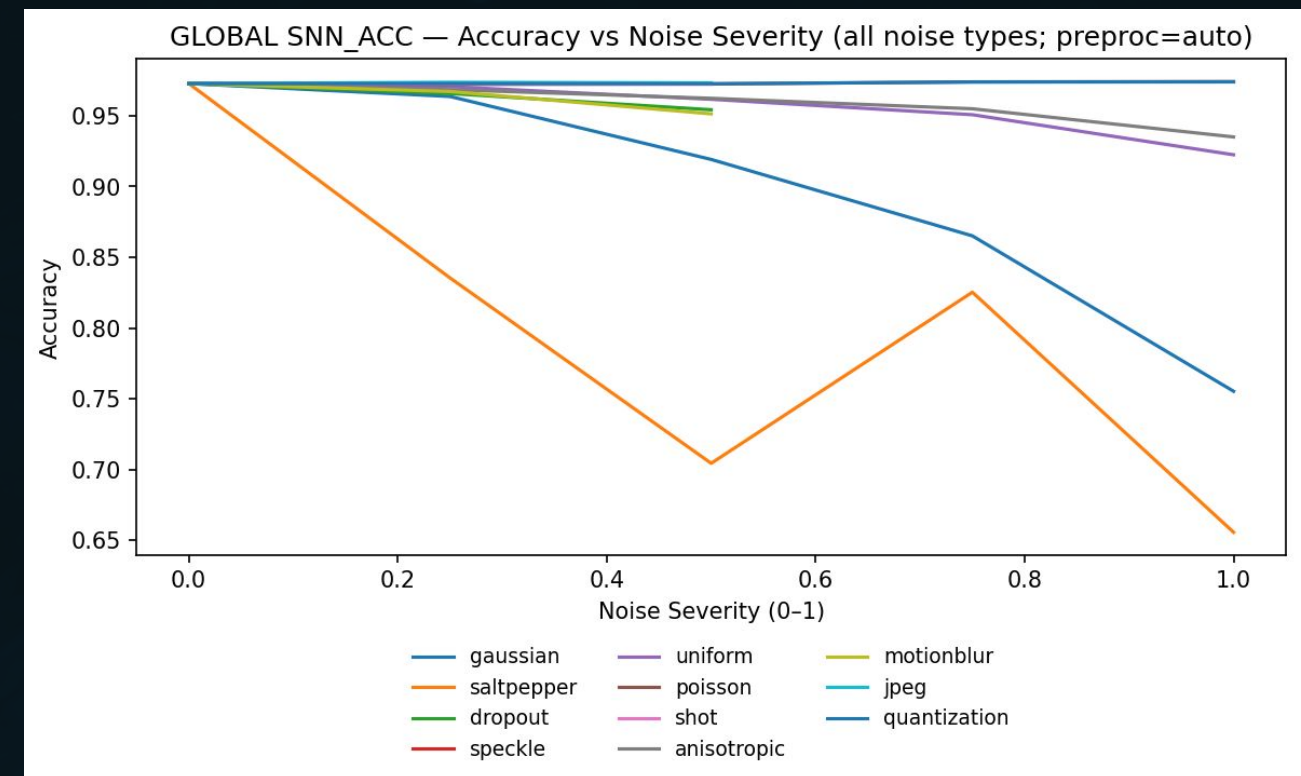
Robustness curves flatten where preprocessing is automatically rejected by the validation gate

Solution: Train-Time Alignment

Fuzzy with alignment



SNN with alignment

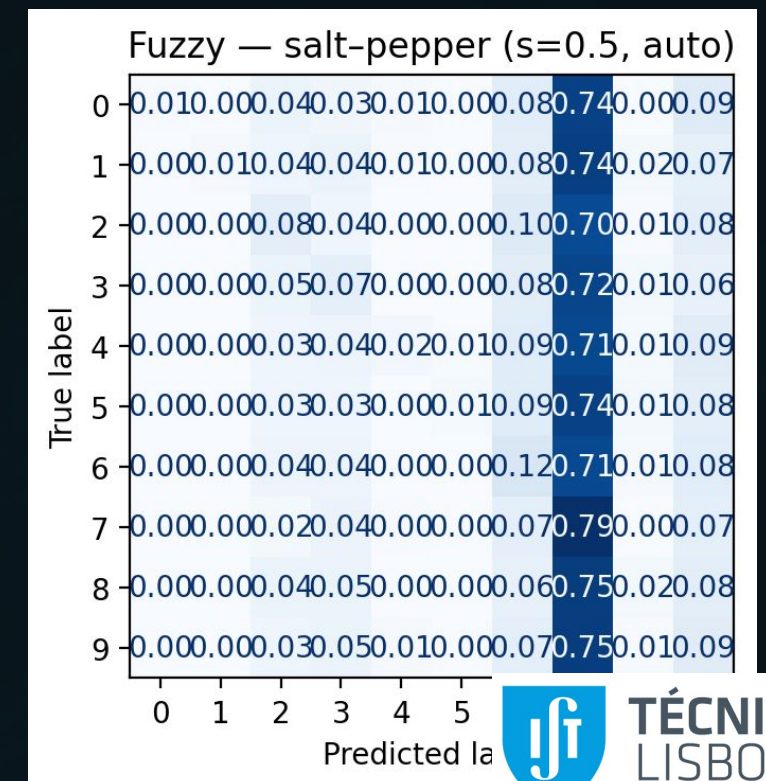
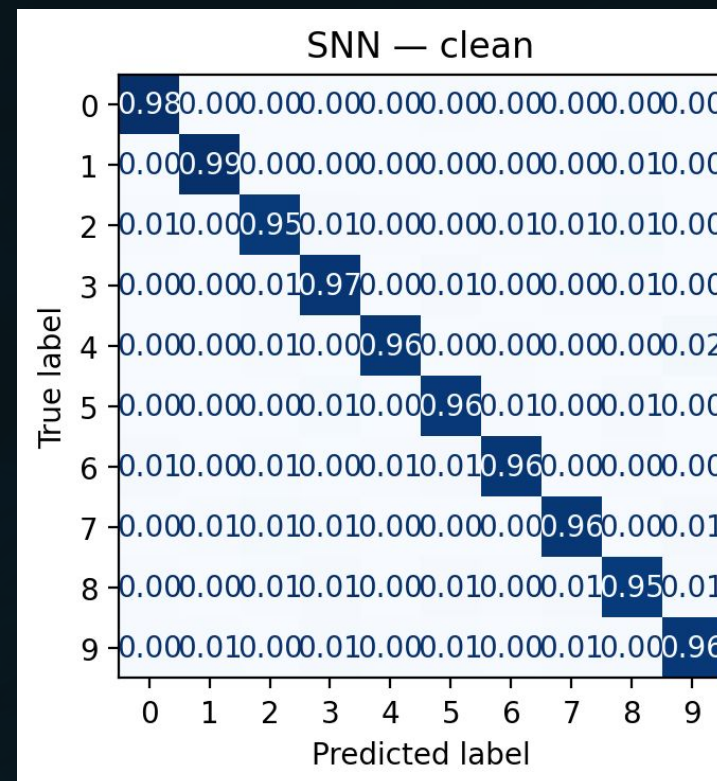
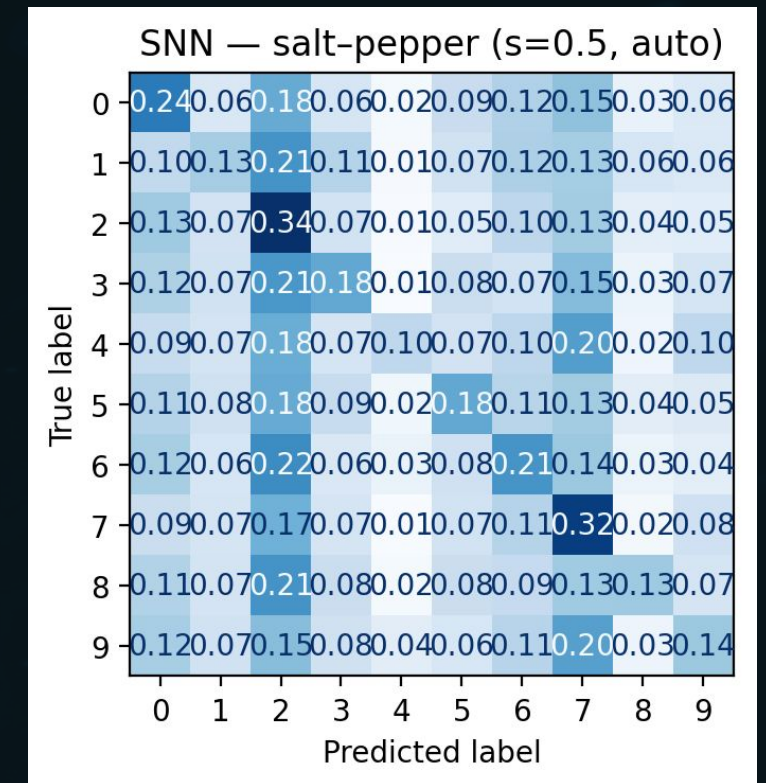
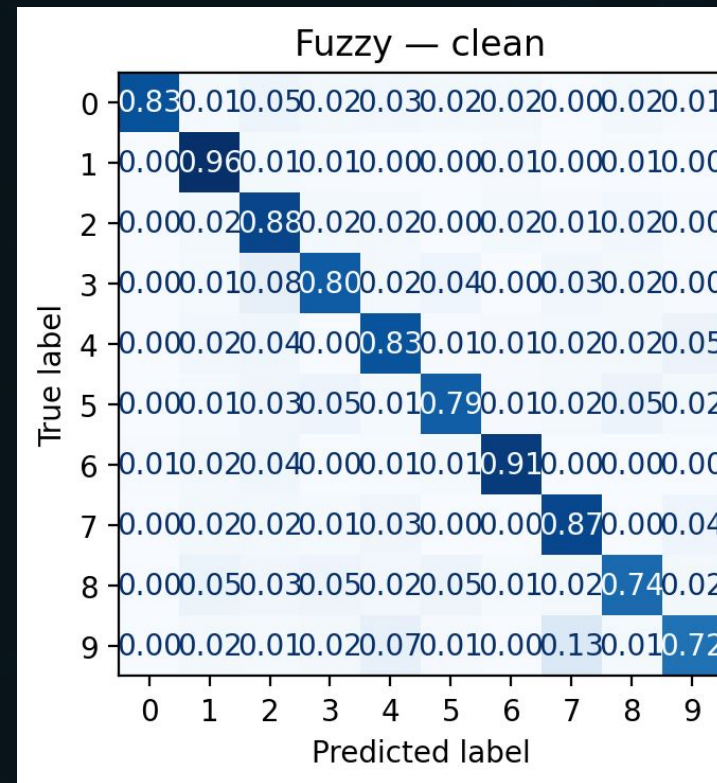


Confusion Matrices

Fuzzy – mode collapse under salt & pepper (maps digits to one class);

SNN – structured confusion among look-alikes rather than collapse;

Preprocessing helps only when the model can exploit the smoothed signal.



Output tables

Table 1: Accuracy (%) by noise type and severity.

Model	Preproc	$s=0.0$	$s=0.25$	$s=0.5$	$s=0.75$	$s=1.0$
<i>Gaussian</i>						
SNN	none	97.33	95.69	90.92	80.35	67.02
Fuzzy	none	86.90	70.84	32.89	16.45	12.41
SNN	gaussian blur	—	—	91.89	83.12	70.55
Fuzzy	gaussian blur	—	—	43.20	25.39	17.76
<i>Salt & Pepper</i>						
SNN	none	97.33	79.81	64.04	50.62	40.19
Fuzzy	none	86.90	17.87	12.68	11.47	11.35
SNN	median	—	—	86.06	82.12	63.57
Fuzzy	median	—	—	26.05	36.04	16.78
<i>Dropout</i>						
SNN	none	97.33	96.38	95.27	93.38	90.91
Fuzzy	none	86.90	86.72	84.84	80.17	69.85
SNN	NL-Means	—	—	24.33	22.93	21.28
Fuzzy	NL-Means	—	—	14.26	13.58	11.87
<i>Speckle</i>						
SNN	none	97.33	97.21	97.09	96.94	96.83
Fuzzy	none	86.90	87.33	87.68	87.28	87.26
SNN	TV	—	—	94.63	94.39	93.83
Fuzzy	TV	—	—	71.09	71.69	70.77
<i>Poisson</i>						
SNN	none	97.33	97.29	97.27	97.28	97.31
Fuzzy	none	86.90	87.09	87.13	87.11	87.08
SNN	wavelet	—	—	97.21	97.31	97.28
Fuzzy	wavelet	—	—	87.73	87.81	87.76

Table 2: Accuracy (%) by noise type and severity.

Model	Preproc	$s=0.0$	$s=0.25$	$s=0.5$	$s=0.75$	$s=1.0$
<i>Uniform</i>						
SNN	none	97.33	96.90	95.93	94.08	90.77
Fuzzy	none	86.90	84.45	72.18	49.33	29.48
SNN	Gaussian Blur	—	—	95.96	94.34	90.83
Fuzzy	Gaussian Blur	—	—	74.46	53.58	32.50
<i>Motion Blur</i>						
SNN	none	97.33	96.68	94.80	92.08	92.08
Fuzzy	none	86.90	84.92	78.54	70.80	70.80
SNN	Wiener	—	—	70.22	45.08	45.08
Fuzzy	Wiener	—	—	47.42	29.95	29.95
<i>Jpeg Compression</i>						
SNN	none	97.33	97.24	97.30	97.23	97.19
Fuzzy	none	86.90	87.10	87.12	87.21	86.82
SNN	NL-Means	—	—	20.99	19.48	19.42
Fuzzy	NL-Means	—	—	12.60	12.56	12.57
<i>Shot</i>						
SNN	none	97.33	97.22	97.23	97.09	97.18
Fuzzy	none	86.90	87.15	87.29	87.40	87.32
SNN	Wavelet	—	—	97.21	97.17	97.13
Fuzzy	Wavelet	—	—	87.99	88.05	87.91
<i>Quantization</i>						
SNN	none	97.33	97.33	97.31	97.38	97.27
Fuzzy	none	86.90	86.86	86.84	86.85	86.68
SNN	median	—	—	97.26	97.24	97.23
Fuzzy	median	—	—	87.75	88.15	88.07
<i>Anisotropic</i>						
SNN	none	97.33	96.71	95.99	94.65	92.63
Fuzzy	none	86.90	83.39	73.05	55.28	37.44
SNN	Gaussian Blur	—	—	96.02	94.72	92.94
Fuzzy	Gaussian Blur	—	—	75.32	58.75	41.53

Answering Our Research Questions

How do models degrade?

SNNs fail gradually through distributed degradation. Fuzzy models collapse catastrophically when structural assumptions break, especially under discontinuous noise.

Does preprocessing help?

Only when: (1) the model architecture can exploit recovered structure, and (2) identical filtering appears during training to prevent distribution mismatch. Train-test alignment is critical.